

DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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Sales Item Management - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Sales Item Management project is a front-end application governed by a strict technical constitution. It implements a decoupled architecture based on a strict separation of concerns between semantic HTML, modular CSS, and ES6+ JavaScript. The project follows a mobile-first responsive pattern and a linear development workflow (HTML -> CSS -> JS -> Validation) to ensure accessibility and maintainability.

1.2 Maturity Assessment

The project specifications are structurally sound regarding coding standards but are currently in a state of REFINEMENT. While the core principles are well-defined, there is a high-severity gap concerning the absence of a formal Testing Discipline & Gates section, leaving the 'Validation' phase of the workflow without concrete technical thresholds or CI/CD requirements.

1.3 Technical Stack

- HTML
- CSS
- JavaScript ES6+

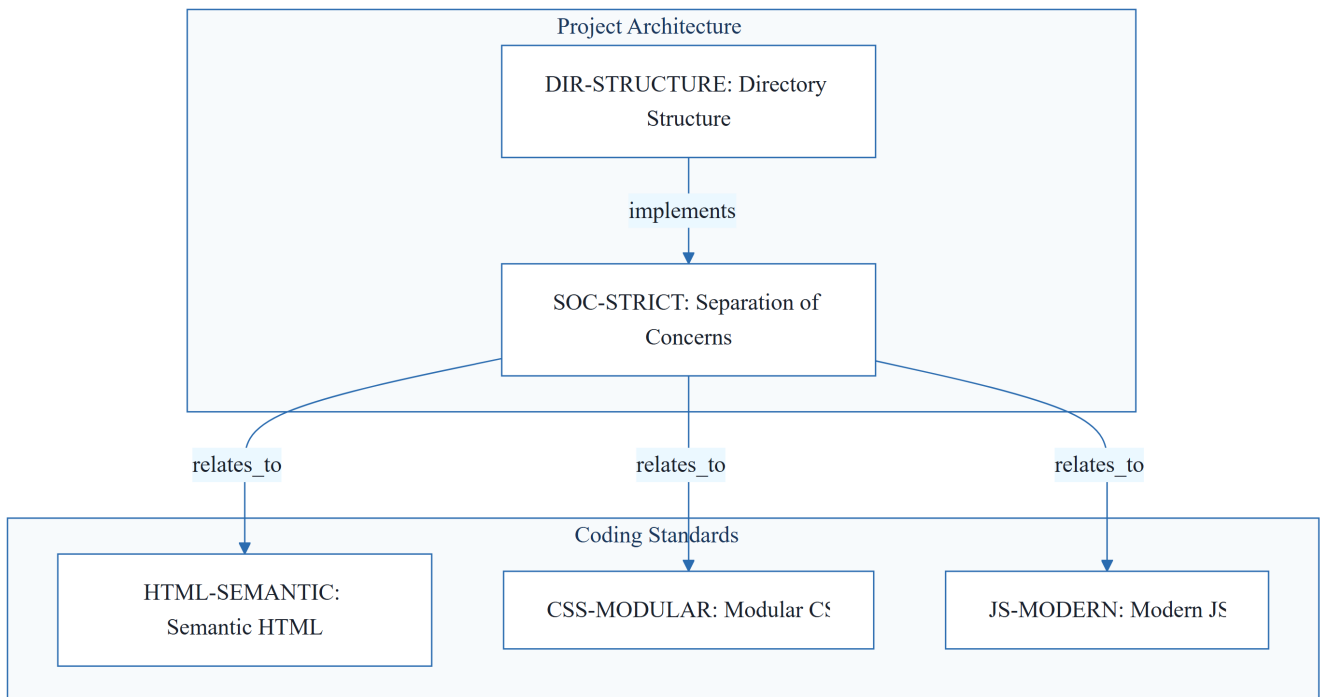
1.4 Architectural Constraints

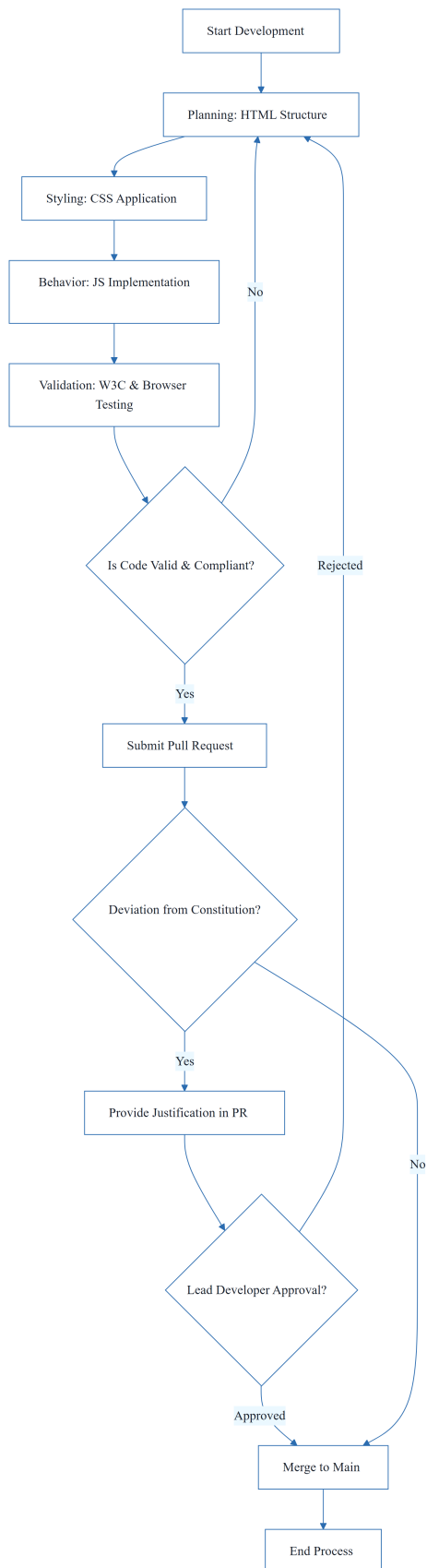
- W3C semantic validity for HTML.
- Mandatory alt attributes for images and labels for forms.
- BEM naming convention for CSS.
- CSS Variables for colors, spacing, and typography.
- Flexbox and Grid for layout responsiveness.
- Mobile-first responsive design strategy.
- Strict prohibition of inline styles and inline event handlers.
- Async/await with try/catch for all asynchronous operations.
- Fixed directory structure: /index.html, /css/, /js/ (with /modules), /assets/.
- Mandatory lead developer approval for any constitutional deviation in PRs.

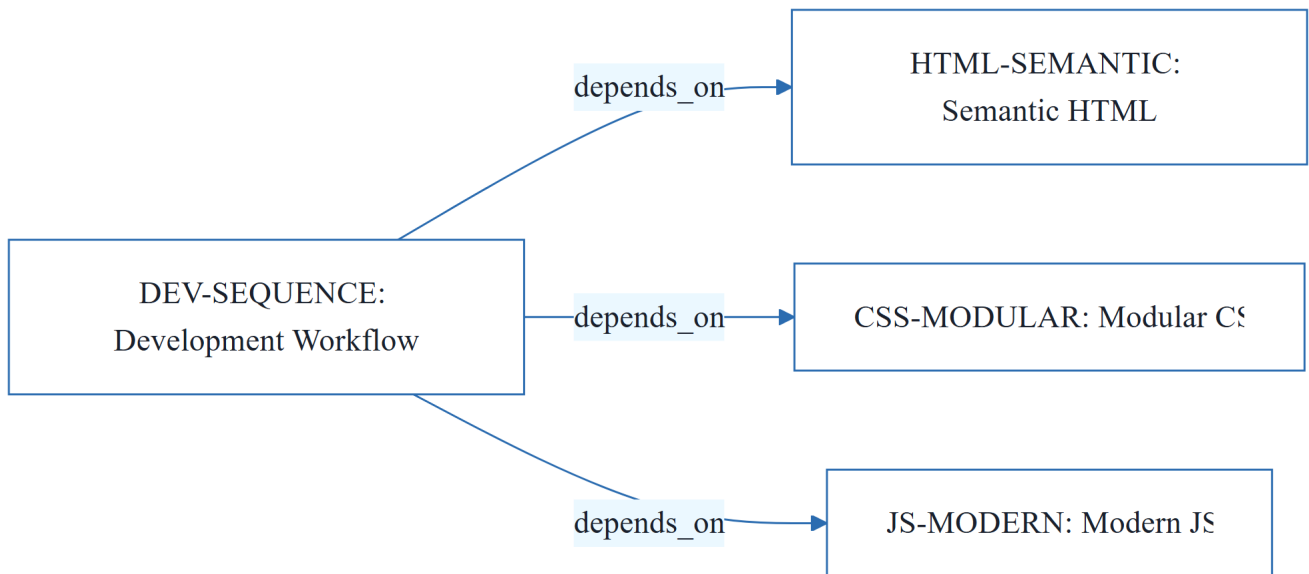
1.5 Critical Dependencies

- W3C Validation tools for HTML/CSS compliance.
- Browser compatibility testing for JS functionality.
- Strict sequential dependency: HTML structure must be defined before CSS styling and JS behavior implementation.
- Lead developer sign-off as a non-negotiable workflow gate for PR merges.

2. Architecture Workflows



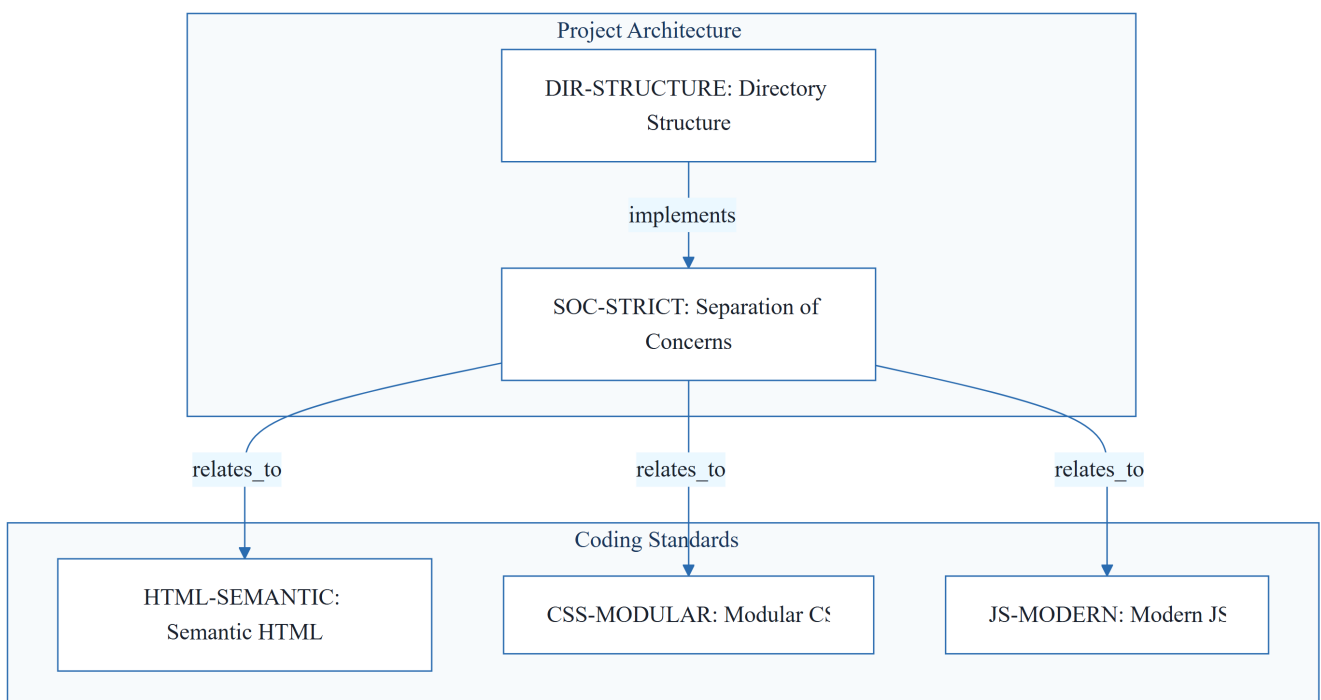




& Visual Diagrams

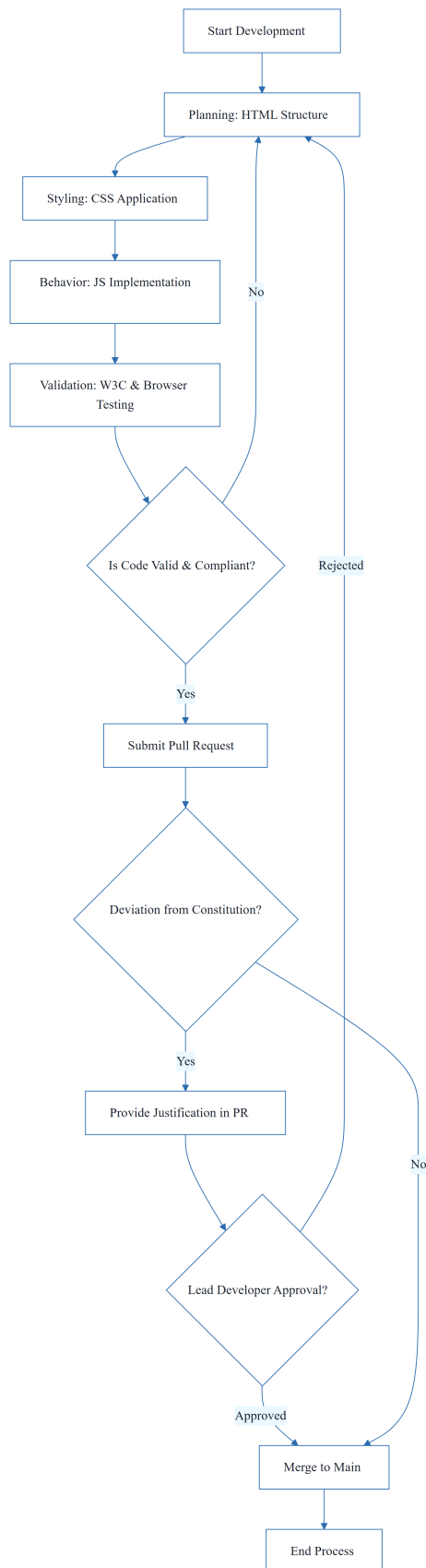
2.1 Technical Standards Traceability Map

Maps the relationships between project structure, separation of concerns, and specific coding standards.



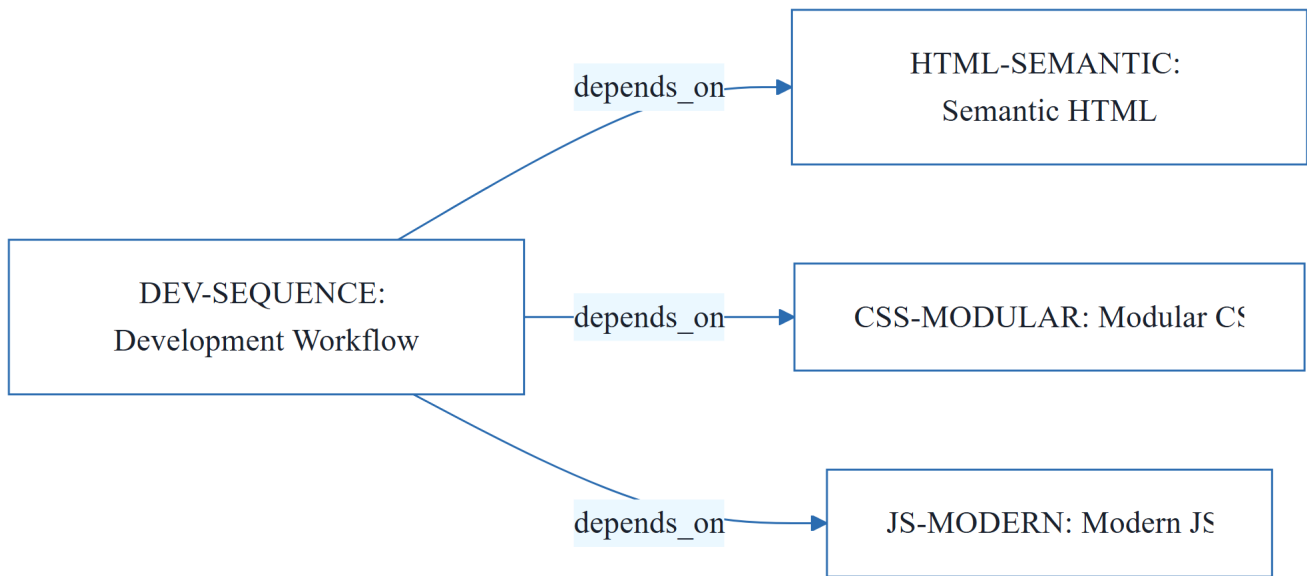
2.2 Development Workflow Process

Detailed development sequence including the mandatory validation and governance check loop.



2.3 Development Dependency Flow

Visualizes how the development sequence depends on the established technical standards.



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Type	Requirement Description	Source Section
HTML-SEMANTIC	coding_standard	HTML must be semantic, W3C valid, with mandatory alt attributes for images and labels for forms.	I. Semantic & Valid HTML
CSS-MODULAR	coding_standard	CSS must use a modular approach (BEM), CSS Variables for consistency, and Flexbox/Grid for layouts.	II. Modular & Maintainable CSS
JS-MODERN	coding_standard	JavaScript must use ES6+ syntax, be modular, and handle async operations via async/await with try/catch.	III. Clean & Modern JavaScript (ES6+)
SOC-STRICT	rule	Strict separation of HTML, CSS, and JS; no inline styles or inline event handlers permitted.	IV. Separation of Concerns
RESPONSIVE-FIRST	requirement	Application must be fully responsive using a mobile-first approach with media queries.	V. Responsive-First Design

Identifier	Type	Requirement Description	Source Section
DIR-STRUCTURE	tool_configuration	Directory structure: /index.html, /css/, /js/ (with /modules), and /assets/.	Project Structure
DEV-SEQUENCE	workflow_constraint	Development sequence: Planning (HTML) -> Styling (CSS) -> Behavior (JS) -> Validation.	Development Workflow
GOV-PR-APPROVAL	rule	Any deviation from the constitution must be justified in the PR and approved by the lead developer.	Governance

3.2 Security Rules

- **Separation of Concerns (SOC-STRICT):** Prohibition of inline event handlers (e.g., `onClick`) to mitigate XSS risks and maintain a clean security boundary between markup and logic.

3.3 Data Models

- **Directory Structure (DIR-STRUCTURE):**
- `/index.html`: Entry point.
- `/css/`: Stylesheets (`style.css`, `variables.css`).
- `/js/`: Logic (`main.js` and `/modules` for components).
- `/assets/`: Static assets (images, icons, fonts).

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Gap	Priority	Remediation Advice
Testing Discipline & Gates	HIGH	The document mentions 'Validation' in the workflow but lacks specific testing gates (e.g., Unit tests, Integration tests, CI/CD pipeline requirements).
Open Questions & Uncertainties	LOW	No known uncertainties are listed; consider adding a section for future technical debt or architectural questions.

4.2 Remediation & Workflow

Deviations from the established constitution must be documented within the Pull Request (PR) and require explicit approval from the Lead Developer. Any amendments to the constitution itself

require a version bump and a documented rationale.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
BEM	TECHNICAL_STACK	CSS-MODULAR	A modular naming convention used to prevent specificity conflicts and improve maintainability.
Behavior	BUSINESS_DOMAIN	DEV-SEQUENCE	The third phase of the development sequence focusing on implementing functional logic.
CSS	TECHNICAL_STACK	CSS-MODULAR	The presentation layer governed by modularity, variables, and responsive layout engines.
HTML	TECHNICAL_STACK	HTML-SEMANTIC	The structural layer required to be semantic and compliant with W3C standards for accessibility.
JS	TECHNICAL_STACK	JS-MODERN	The logic layer utilizing ES6+ syntax and asynchronous patterns.
JavaScript	TECHNICAL_STACK	JS-MODERN	The logic layer utilizing ES6+ syntax and asynchronous patterns.
PR	TECHNICAL_STACK	GOV-PR-APPROVAL	The mechanism where deviations from the constitution must be justified and approved by the lead developer.
Planning	BUSINESS_DOMAIN	DEV-SEQUENCE	The initial phase of the development sequence dedicated to defining the structural markup.
SEO	TECHNICAL_STACK	HTML-SEMANTIC	The optimization goal achieved through the use of semantic tags and valid markup.
Styling	BUSINESS_DOMAIN	DEV-SEQUENCE	The second phase of the development sequence where design requirements are applied to the markup.

Term	Category	Context Anchor	Project Definition
Validation	BUSINESS_DOMAIN	DEV-SEQUENCE	The final phase of the development sequence involving official checkers and cross-browser testing.